

Dr Shubham Tiwari

Postdoctoral Research, Jornada Experimental Range & NMSU

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Postdoctoral researcher at New Mexico State University and the Jornada Experimental Range, studying dust source dynamics and dryland connectivity in the American Southwest. Holds a PhD from Durham University (2025) and previously served as Senior Hydrologist at the UK Environment Agency, leading catchment hydrology, drought monitoring and permitting across Hertfordshire and North London..

Employment

- 2026– **Postdoctoral Researcher**, New Mexico State University & Jornada Experimental Range.
- 2024–2026 **Senior Hydrologist**, Hertfordshire and North London, Environment Agency, UK.
- 2020–2023 **Marie-Curie Early Stage Researcher**, i-CONN project, Durham University, UK.
- 2022 **Visiting Researcher**, Department of Geosciences, University of Arizona, United States.
- 2021 **Visiting Researcher**, Inst. of Hydrobiology and Aquatic Ecosystem Man., BOKU, Austria.
- 2019–2020 **Junior Research Fellow**, Science and Engineering Board, DST, India.
- 2018–2019 **Research Assistant**, Department of Earth and Env. Sciences, IISER Bhopal, India.
- 2017–2018 **Research Intern**, Department of Mathematics, MANIT Bhopal, India.

Education

- 2020–2025 **Durham University, United Kingdom**, Ph.D. Physical Geography.
Thesis – *Patterns of connectivity in ecogeomorphic systems: a network-based approach to understanding land degradation in drylands*
Supervisors: Dr Laura Turnbull-Lloyd and Prof John Wainwright
- 2018–2019 **Indian Institute of Science Education and Research Bhopal, India**, MS Thesis.
Thesis – *Application of network theory to reconstruct daily rainfall data in the Murray Darling basin*
Supervisor: Dr Sanjeev Kumar Jha
- 2014–2019 **Indian Institute of Science Education and Research Bhopal, India**, BSMS Dual Degree (in Earth and Environmental Sciences).

Selected Awards and Fellowships

- 2023 British Geomorphological Society Travel Grant – United Kingdom (£500)
- 2022 Best Start-Up Pitch Award, "PhD to Entrepreneur" Jacobs University – Germany
- 2020–2023 Marie Skłodowska-Curie Doctoral Scholarship (i-CONN) – UK (approx. £230,000)
- 2020–2023 MEXT Scholarship – Japan (approx. £120,000 for doctoral studies; declined)
- 2014–2019 Innovation in Science Pursuit for Inspired Research (INSPIRE) Fellowship, Department of Science and Technology – India (approx. £7,500)
- 2017 Nominated for Indian Youth Exchange Program to China – India

Professional Service

- 2025–Present Peer reviewer: Journal of Hydrology; Ecohydrology; ESPL; Scientific Reports.
- 2024–Present Co-chair, National Hydrology Outreach Group – Environment Agency
- 2024–Present Data Management Lead, Groundwater & Hydrology Team – Environment Agency
- 2024 Acting Team Manager (cover), Groundwater & Hydrology Team, Environment Agency
- 2024–2025 EDI Committee Representative, British Geomorphological Society
- 2020–2023 Early Stage Researcher Representative, i-CONN Working Group "Creating transdisciplinary capacity and uptake of research outcomes"

- 2022 School outreach organiser, Peases West Primary School – “How to Become a Scientist”.
- 2021 Volunteer, ECO2 Schools Climate Conference, Durham
- 2017–2019 Lawn Tennis Club Coordinator, sports council at IISER Bhopal.
- 2017–2018 Class Representative, batch of BSMS 2014 at IISER Bhopal.

Selected Publications

Peer-Reviewed Publications

1. **Tiwari, S.**, Turnbull, L., & Wainwright, J. (2025). “Local- and global-scale hydrological and sediment connectivity over grassland and shrubland hillslopes”, *Journal of Hydrology*, 655, 132896. [link here](#)
2. **Tiwari, S.**, Turnbull, L., & Wainwright, J. (2024). “Water-controlled ecosystems as complex networks: Evaluation of network-based approaches to quantify patterns of connectivity”, *Ecohydrology*, 17(7), e2690. [link here](#)
3. Karittevlis, C., Papadopoulos, M., Lima, V., Orphanides, G.A., **Tiwari, S.**, Antonakakis, M., Papadopoulou Lesta, V., & Ioannides, A.A. (2024). “First activity and interactions in thalamus and cortex using raw single-trial EEG and MEG elicited by somatosensory stimulation”, *Frontiers in Systems Neuroscience*, 17, 1305022. [link here](#)
4. Recinos, S., Funk, A., **Tiwari, S.**, Baldan, D., & Hein, T. (2024). “Multilayer Networks in Landscape Ecology: A Case Study to assess changes in aquatic habitat connectivity for flying and non-flying benthic macroinvertebrates in a Danube floodplain”, *Landscape Ecology*, 39, 186. [link here](#)
5. Singh, A., **Tiwari, S.**, Jha, S. K. (2021). “Evaluation of quantitative precipitation forecast in five Indian river basins”, *Hydrological Sciences Journal*, 66(15), p. 2216–2231. [link here](#)
6. Vousta, V. and others (2021). “Two classes of functional connectivity in dynamical processes in networks”, *Journal of the Royal Society Interface*, 18(183). [link here](#)
7. **Tiwari, S.**, Jha, S. K., & Singh, A. (2020). “Quantification of node importance in rain gauge network: influence of temporal resolution and rain gauge density”, *Scientific Reports*, 10(1), p. 9761. [link here](#)
8. **Tiwari, S.**, Jha, S. K., & Sivakumar, B. (2019). “Reconstruction of daily rainfall data using the concepts of networks: accounting for spatial connections in neighborhood selection”, *Journal of Hydrology*, 579. [link here](#)

Manuscripts Under Review

1. **Tiwari, S.** et al. (2025). “The role of connectivity in understanding grassland-shrubland regime shifts: impacts of aridity, grazing and wind dynamics”, under review in *Ecological Modelling*.
2. **Tiwari, S.**, Turnbull, L., & Wainwright, J. (2025). “From theory to action: Using connectivity to inform adaptive grazing in drylands”, under review in *Earth’s Future*. Manuscript Id: 2025EF007531.
3. Ali, S. H.; **Tiwari, S.** et al. (2025). “Understanding patterns of soil erosion: a minimal model approach”, under review in *Royal Society Interface*.
4. **Tiwari, S.**, Turnbull, L., & Wainwright, J. “Reorganisation of UK daily rainfall connectivity between 1961 and 2020: A complex network analysis”, under review in *Journal of Hydrology*.

Note: I have received extensive training in various aspects of a career in academia and industry through the i-CONN project, including workshops on academic writing, project management, lightning talks, entrepreneurship, dealing with media and conflict resolution. More information can be found [here](#).